

SOUPAL PURKAYASTHA

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EDUCATION

Jadavpur University, Kolkata

Expected Graduation: May 2028

B.Tech, Information Technology | CGPA: 8.93/10

Relevant Coursework: Data Structures & Algorithms, OOP, DBMS, Computer Networks, Operating Systems

Delhi Public School Joka (South), Kolkata

Jun 2022 – Apr 2024

Higher Secondary — 92.2%

Delhi Public School Joka (South), Kolkata

Apr 2021 – May 2022

Secondary School Certificate — 96%

PROJECTS

SnapLink — URL Shortener with Security-Hardened Link Analytics

[\[Github\]](#) [\[Live Demo\]](#)

- Built and deployed a full-stack URL shortener with JWT authentication, guest and logged-in flows, and a Recharts analytics dashboard tracking device, browser, and referrer data per link.
- Hardened the link-creation endpoint against SSRF and phishing by resolving hostnames via DNS and blocking private/loopback IP ranges, and integrating Google Safe Browsing API to reject malicious URLs before shortening.
- Implemented dual-tier rate limiting (per-IP request caps on link creation vs. general API traffic) and MongoDB TTL indexing for automatic link expiration, reducing abuse risk and stale data.
- Defended against NoSQL injection, brute-force login, and race conditions in click tracking using atomic MongoDB operations and constant-time password verification.

Technologies: React, Node.js, Express, MongoDB, JWT, Recharts

FinTrack — Client-Side Personal Finance Ledger

[\[Github\]](#) [\[Live Demo\]](#)

- Built a privacy-first finance tracker with zero external network calls or backend dependency — all transaction data lives in client-side React state, with categorized income/expense logging and real-time balance calculation via functional array reduction.
- Implemented input validation guardrails rejecting empty, zero, negative, and NaN transaction amounts before they reach state, preventing invalid ledger entries.
- Designed an interactive category-breakdown visualization using Recharts (pie chart with dynamic color mapping and custom tooltip formatting) that recalculates from raw transactions on every state change.
- Built a fully responsive CSS Grid layout with a custom ledger-themed design system using CSS custom properties.

Technologies: React, JavaScript (ES6+), Recharts, Vite, CSS3

CODING PROFILES

LeetCode: 700+ problems solved | Knight | Max Rating: 1923 [\[Profile\]](#)

CodeForces: Pupil | Max Rating: 1240 [\[Profile\]](#)

CodeChef: 3-Star Coder | Max Rating: 1608 [\[Profile\]](#)

ACHIEVEMENTS

- Secured Global Rank 430 in CodeChef Starters 232 among 4,328 participants
- Secured Rank 999 in LeetCode Weekly Contest 491 among 40,616 participants
- Solved 1,300+ algorithmic problems across LeetCode, CodeChef, Codeforces, and GeeksforGeeks
- Secured AIR 9278 in JEE Advanced 2024 (99.23 percentile in JEE Mains 2024) and GMR 412 in WBJEE 2024